E-Learning

Software Requirements Specification

Version 1.0

Group Id: S2302E3649 (BC200202841)

Supervisor Name: Maham Murtaza Khan

Revision History

Date (dd/mm/yyyy)	Version	Description	Author
27/6/2023	1.0	This document will contain the introductory lines regarding my project. It will contain the scope of the project that is being developed. Scope of the project will be explained on the basis of requirements (Functional and Non-Functional) are related to project that will be developed. A Use Case Diagram is also present in this document. It explains that how a user will act and work on this software. Its complete steps are given in this document and usage scenarios explain that how a Use will work. At the end, we also explained working methodology and work flow chart.	**(BC200202841)**

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SRS Document

Scope of Project:

The scope of the E-Learning project is to develop a self-paced Learning Management System (LMS) that allows users to access and enroll in various courses. The project falls under the domain of web programming and aims to provide a platform for users to enhance their knowledge and skills through online courses.

The system consists of two main panels: the Admin Panel and the User Panel. The Admin Panel allows the administrator to manage the courses and user registrations, while the User Panel provides the interface for users to browse, enroll in, and complete courses.

In the Admin Panel, the administrator has the ability to log in and out of the system. They can also add additional administrators to assist with managing the platform. The admin has the responsibility to add course information, including the course title, instructor details, course content, videos, reading materials, outline, and quizzes. Each type of information has a separate interface or page for easy management. The admin can offer courses for free or set a price for paid courses. They can also generate and send coupons to users through email. Additionally, the admin can review and accept/reject user registration requests, view the number of students registered for each course, and respond to user queries.

In the User Panel, users have the option to view all available courses on the home page without logging in. However, to enroll in a course, users must first register on the website. After registration, users can log in and log out as needed. Users can access detailed information about each course, including the course title, instructor details, content, and outline. They can also ask questions and engage in discussions through the comment section.

Users can enroll in courses and complete them at their own pace. To complete a course, users must watch a certain percentage (e.g., 80%) of the videos in each section. The system also includes quizzes based on multiple-choice questions (MCQs) to assess the users' knowledge. Users can attempt the quizzes and view their results. If they score less than 75% on a quiz, they have the option to retake it. The platform provides various filters for users to search for courses based on their preferences, such as free/paid courses or specific topics.

The system also incorporates features for users to receive and apply coupons for paid courses. Users can receive coupons through email and use them to access discounted or free courses. The platform keeps track of users' attempted courses and provides an attempt history for reference. Upon completion of a course, users can download a PDF certificate with their name included.

Functional and non Functional Requirements:

Functional Requirements:

1. User Registration: The system should allow users to register an account by providing necessary details such as name, email, and password.
2. User Login and Logout: Users should be able to securely log in and log out of their accounts to access the system.
3. Course Browsing: Users should be able to browse through the available courses on the website without the need for login. They should be able to view course details, including the course title, instructor information, content, and outline.
4. Course Enrollment: Users should be able to enroll in courses of their choice. Upon enrollment, users gain access to the course materials and resources.
5. Course Completion Tracking: The system should track the progress of users within each course, such as tracking the percentage of videos watched and sections completed.
6. Course Assessment: The system should provide quizzes based on multiple-choice questions (MCQs) to assess the users' understanding of the course material.
7. Certificate Generation: Upon successful completion of a course, users should be able to download a PDF certificate with their name included.

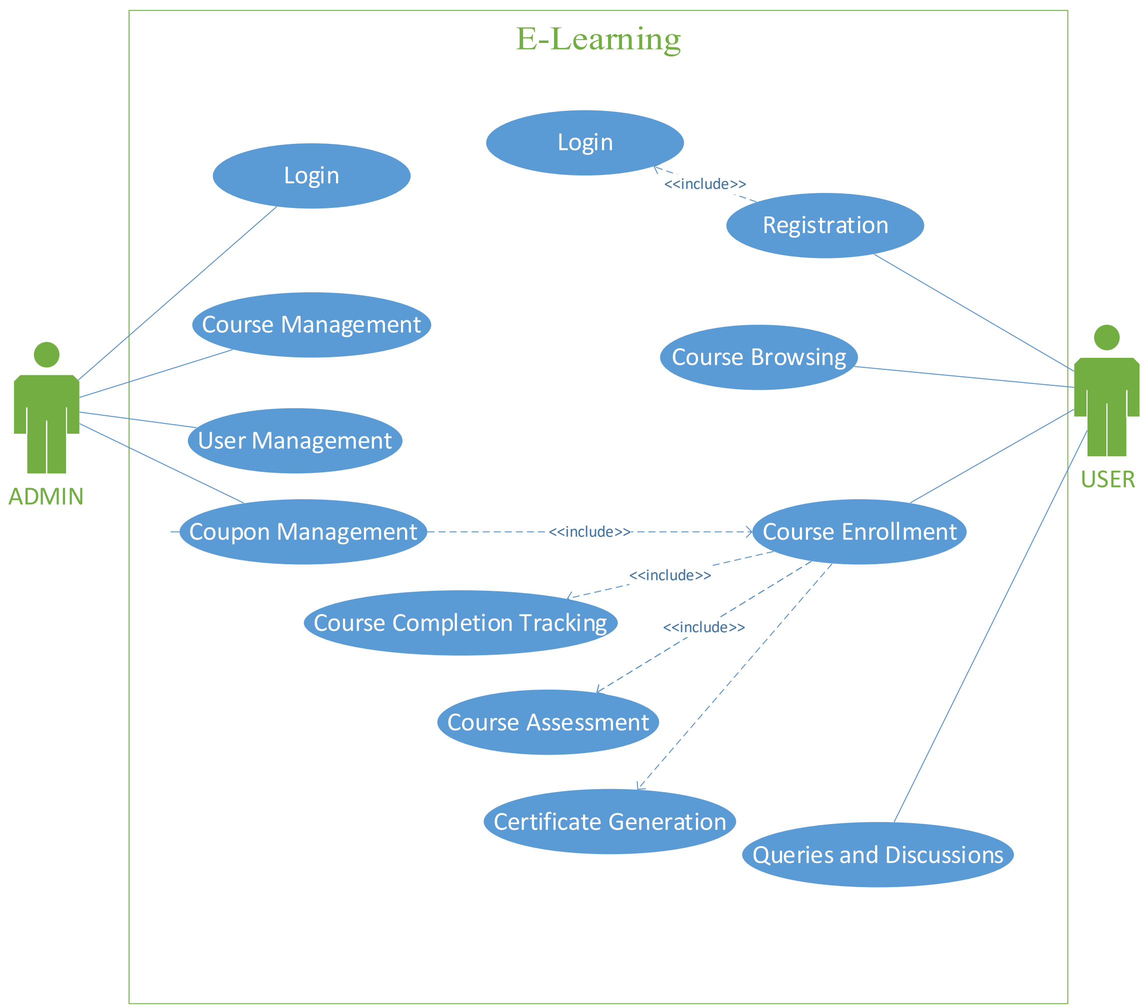
8. User Queries and Discussions: Users should have the ability to ask questions and participate in discussions related to specific courses through a comment section or discussion forum.
9. Admin Login: The system should provide a secure login for administrators to access the admin panel.
10. Course Management by Admin: The admin should have the capability to add new courses, manage course content (including videos, reading materials, and quizzes), and set the course price.
11. User Management by Admin: The admin should be able to view and manage user registrations, accept/reject registration requests, and respond to user queries.
12. Coupon Management: The admin should be able to generate and distribute coupons for paid courses. Users should be able to apply these coupons during course enrollment.

Non-Functional Requirements:

1. Usability: The system should have a user-friendly interface with intuitive navigation and clear instructions for easy usage by users of varying technical abilities.
2. Security: The system should implement proper security measures, including encryption of user data, secure user authentication, and protection against unauthorized access or data breaches.
3. Performance: The system should be able to handle concurrent user requests and provide a responsive experience, ensuring minimal downtime and fast page loading times.
4. Scalability: The system should be designed to accommodate a growing number of users, courses, and course materials without compromising performance.
5. Compatibility: The system should be compatible with various web browsers and accessible across different devices, including desktops, laptops, tablets, and mobile phones.
6. Reliability: The system should be reliable, ensuring that user data is accurately stored and courses can be accessed without any disruptions or data loss.
7. Maintainability: The system should be designed in a modular and well-structured manner, making it easy for administrators to maintain and update course content, user data, and system functionalities.
8. Accessibility: The system should adhere to accessibility standards, allowing users with disabilities to access and use the platform effectively.

9. Data Backup and Recovery: The system should have mechanisms in place for regular data backups and a plan for disaster recovery in case of system failures or data loss.
10. Performance Monitoring: The system should incorporate performance monitoring tools to track system performance, identify bottlenecks, and optimize system resources as needed.

Use Case Diagram(s):



Usage Scenarios:

Use Case Title & ID	Actions	Description	Pre-conditions	Post-conditions	Author	Exceptions
UC-01: User Registration	User fills registration form with required details	The user fills in the registration form with their name, email, and password to create a new account in the e-learning system	None	User account is created, and user is logged in	**(BC200202841)**	Email already exists, Invalid email format
UC-02: User Login	User enters login credentials and submits	The user enters their registered email and password to log in to the system	User must be registered	User is successfully logged in	**(BC200202841)**	Invalid email or password
UC-03: User Logout	User clicks on the logout button	The user initiates the logout process to securely log out from the system	User must be logged in	User is logged out, session is terminated	**(BC200202841)**	None
UC-04: Course Browsing	User views the list of available courses	The user can browse and view the list of available courses on the website without logging in	None	User can view course details and enroll in a course	**(BC200202841)**	None
UC-05: Course Enrollment	User selects a course and clicks on "Enroll"	The user selects a course from the list and enrolls in the selected course	User must be logged in	User is enrolled in the course and gains access to course materials	**(BC200202841)**	None
UC-06: Course Completion Tracking	System tracks user progress within a course	The system tracks and updates the user's progress within the enrolled course based on the completion of videos and sections	User must be enrolled in a course	User's course progress is updated	**(BC200202841)**	None
UC-07: Course	User attempts a	The user attempts a quiz to assess	User must be	User's quiz results are	**(BC200202841)**	None

Use Case Title & ID	Actions	Description	Pre-conditions	Post-conditions	Author	Exceptions
Assessment	quiz for a course	their knowledge and understanding of the course material	enrolled in a course	recorded		
UC-08: Certificate Generation	User requests certificate after completing a course	The user requests a certificate upon successful completion of a course	User must have completed a course with the required score	Certificate is generated and available for download	**(BC200202841)**	None
UC-09: User Queries and Discussions	User asks a question or participates in course discussions	The user interacts with other users and instructors by asking questions or engaging in discussions related to a course	User must be enrolled in a course	User's question or comment is posted and visible to others	**(BC200202841)**	None
UC-10: Admin Login	Admin enters login credentials and submits	The admin enters their registered email and password to log in to the admin panel	Admin must be registered	Admin is successfully logged in	**(BC200202841)**	Invalid email or password
UC-11: Course Management	Admin adds or manages course information	The admin adds or manages course details, including content, materials, and quizzes	Admin must be logged in	Course information is added or updated	**(BC200202841)**	None
UC-12: User Management	Admin manages user registrations and queries	The admin manages user registrations, accepts/rejects registration requests, and responds to user queries	Admin must be logged in	User registrations and queries are managed	**(BC200202841)**	None
UC-13: Coupon Management	Admin generates and distributes course coupons	The admin generates and distributes coupons for paid courses, which can be applied by users	Admin must be logged in	Coupons are generated and distributed to users	**(BC200202841)**	None

Use Case Title & ID	Actions	Description	Pre-conditions	Post-conditions	Author	Exceptions
		during course enrollment				

Adopted Methodology

Available Methodologies:

There are the following methodologies:

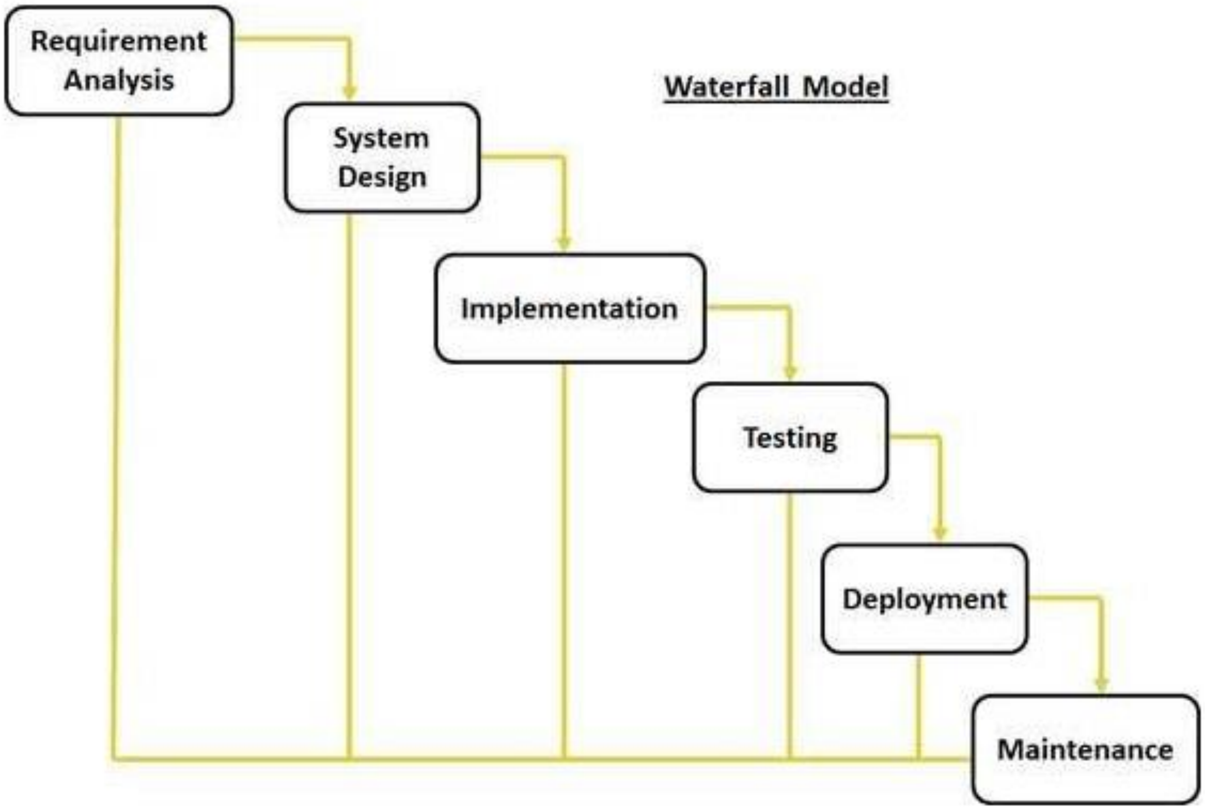
- Build-and-fix model
- Waterfall model
- Rapid prototyping model
- Incremental model
- Extreme programming
- Synchronize-and-stabilize model
- Spiral model
- Object-oriented life-cycle models

CHOSEN METHODOLOGY:

Different methodologies and models are used by the people according to their work and planning. We will use “VU process model” It is combination of two models Water Fall and Spiral methodologies.

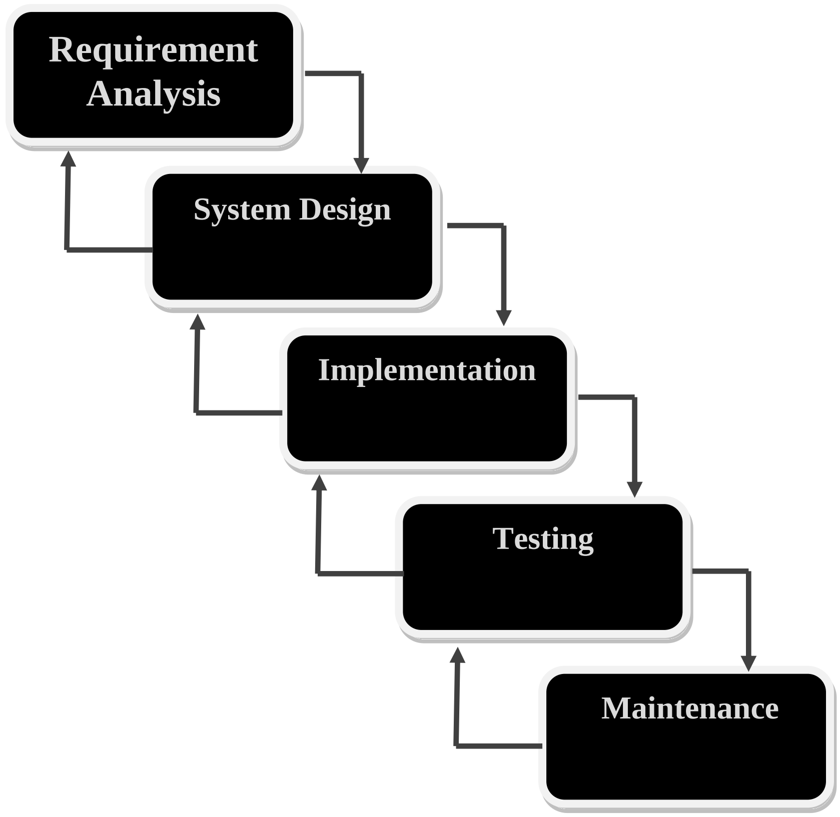
Waterfall Model:

This model is also known as linear sequential model. This model is depicted in the following diagram.



It suggests a systematic, sequential approach to software development that begins at the system level and progresses through the analysis, design, coding, testing, and maintenance.

Real projects rarely follow the sequential flow that the model proposes. In general these phases overlap and feed information to each other. Hence there should be an element of iteration and feedback. A mistake caught any stage should be referred back to the source and all the subsequent stages need to be revisited and corresponding documents should be updated accordingly. This feedback path is shown in the following diagram.

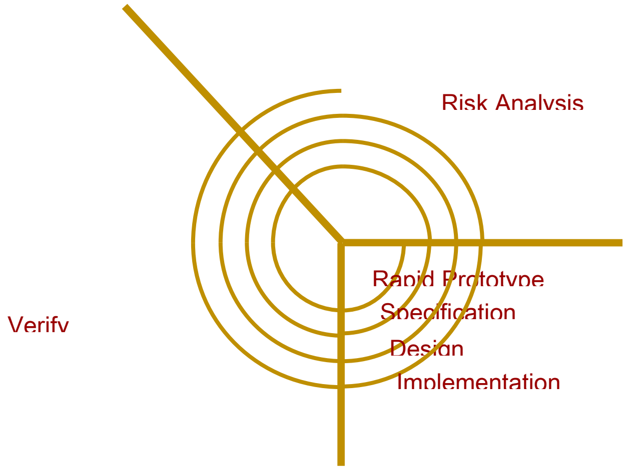


Conclusion:

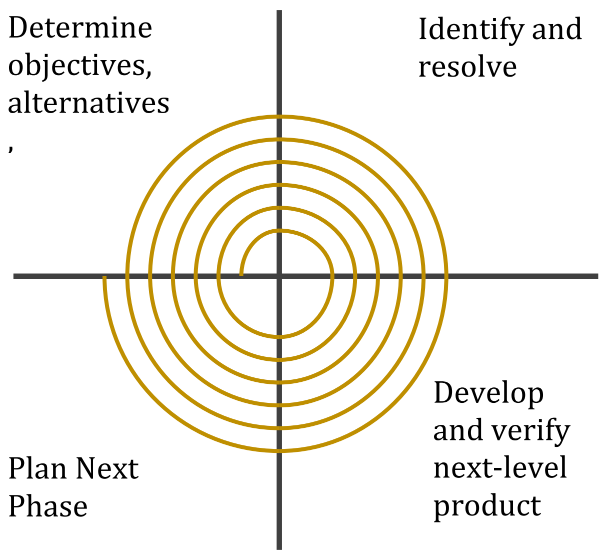
The Waterfall Model is a documentation-driven model. It therefore generates complete and comprehensive documentation and hence makes the maintenance task much easier. It however suffers from the fact that the client feedback is received when the product is finally delivered and hence any errors in the requirement specification are not discovered until the product is sent to the client after completion. This therefore has major time and cost related consequences.

Spiral Model:

This model was developed by Barry Boehm. The main idea of this model is to avert risk as there is always an element of risk in development of software. For example, key personnel may resign at a critical juncture, the manufacturer of the software development may go bankrupt, etc.



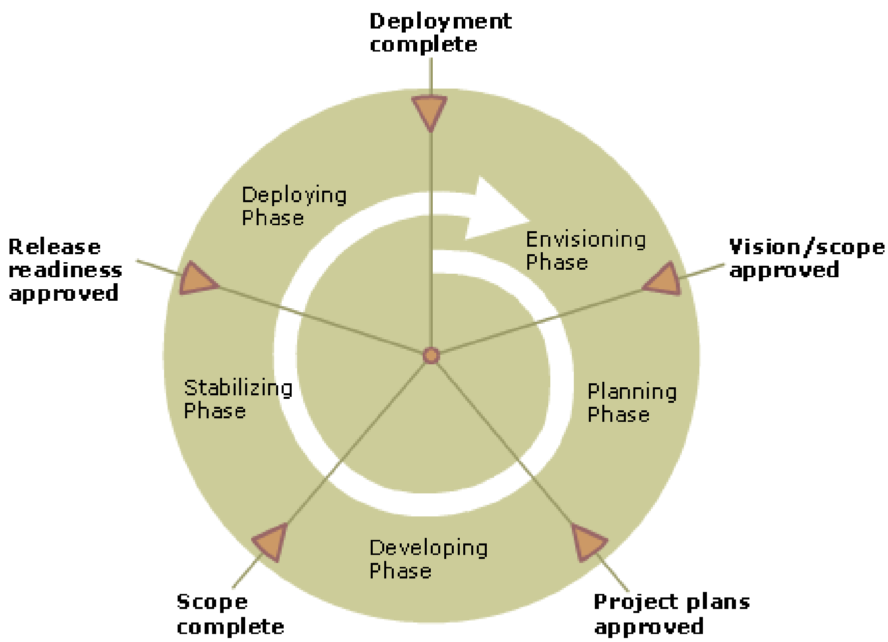
In its simplified form, the Spiral Model is Waterfall model plus risk analysis. In this case each stage is preceded by identification of alternatives and risk analysis and is then followed by evaluation and planning for the next phase. If risks cannot be resolved, project is immediately terminated.



Conclusion:

As can be seen, a Spiral Model has two dimensions. Radial dimension represents the cumulative cost to date and the angular dimension represents the progress through the spiral. Each phase begins by determining objectives of that phase and at each phase a new process model may be followed. The main strength of the Spiral Model comes from the fact that it is very sensitive to the risk. Because of the spiral nature of development, it is easy to judge how much to test and there is no distinction between development and maintenance. It however can only be used for large-scale software development and that too for internal (in-house) software only.

VU Process Model which is the combination of Waterfall and Spiral model. In this model each stage of waterfall is preceded by identification of alternatives and risk analysis and is then followed by evaluation and planning for the next phase.



Reasons for choosing the Methodology

It derives the benefits of predictability from the milestone-based planning of the waterfall model, as well as the benefits of feedback and creativity from the spiral model.

- ✓ The VU Process Model is a documentation-driven model. It therefore generates complete and comprehensive documentation and hence makes the maintenance task much easier because the feedback of user must fulfil on each phase of development.
- ✓ .VU Process Model is heavily dependent on risk analysis and evaluation in each phase.
- ✓ Progress Tracking and Risk management is easy in this process.
- ✓ Project can be divided into small units and thus the Object-Oriented Approach for web based application can easily be implemented.
- ✓ Iterations are easily manageable.
- ✓ Application Quality Assurance can easily be managed by testing the developed modules individually.
- ✓ The modules or prototypes which are successfully developed can be implemented independently and can be easily integrated.
- ✓ The deadline for the given project is good enough to be build. So we can easily choose selected methodology to work upon.

<u>Work Plan (Use MS Project to create Schedule/Work Plan)</u>

ID	Task Name	Start	Finish	Duration
1	SRS	5/31/2023	7/14/2023	33d
2	Design Document	7/17/2023	9/13/2023	43d
3	Prototype Phase	9/14/2023	10/4/2023	15d
4	Final Deliverable	10/5/2023	12/4/2023	43d